

MATHEMATICS FOR ML: SETS, MEASURE & PROOFS

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Outline

① SETS, MEASURE & PROOFS

About Me

Yogesh Haribhau Kulkarni

Bio:

- ▶ 20+ years in CAD/Engineering software development
- ▶ Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- ▶ Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
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About Content

- ▶ Bare essential mathematics needed for Machine/Deep Learning
- ▶ Just to get you started.
- ▶ Each field within Machine/Deep Learning can go extremely deep.
- ▶ This is to give you enough foundation needed.

About Me

- ▶ I am not a mathematician.
- ▶ And this course won't turn YOU into one!!! (if you are not already)
- ▶ But, I will surely attempt to make you get interested in the learning, more.

Sets

Sets

EXAMPLE

The set consisting of all positive integers less than 10 can be denoted by $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. $\{1, 2, 3, \dots, 9\}$ is also used since the general pattern is obvious.

Sets

EXAMPLE

Members of a set need not be numeric. The set of all possible outcomes when tossing a coin is $\{H, T\}$.

Set Builder Notation

For example, $O = \{1, 3, 5, 7, 9\}$ can also be written as

$$O = \{x : x \text{ is a positive odd integer less than } 10\}.$$

EXAMPLE

- ▶ $A = \{x : x \text{ is a positive even number less than } 15\} = \{2, 4, 6, 8, 10, 12, 14\}.$
- ▶ $B = \{n : n \text{ is a positive integer}\} = \{1, 2, 3, 4, 5, \dots\}.$
- ▶ $C = \{2n : n \text{ is a whole number and } 1 \leq n \leq 4\} = \{2, 4, 6, 8\}.$

Some Basic Notations

- ▶ $\emptyset$, The empty set (a set with *no* elements). $\{ \}$ is also used.
- ▶ $\mathbb{N} = \{1, 2, 3, \dots\}$, the set of natural numbers.
- ▶ $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of integers.
- ▶ $\mathbb{R}$ or $(-\infty, \infty)$, the set of real numbers.
- ▶ $[a, b]$, the set of all real numbers x such that $a \leq x \leq b$.
- ▶ (a, b) , the set of all real numbers x such that $a < x < b$.
- ▶ $(a, b]$, the set of all real numbers x such that $a < x \leq b$.
- ▶ $[a, b)$, the set of all real numbers x such that $a \leq x < b$.

(Ref: MAT 1348 B: Discrete Mathematics for Computing - Supartha Podder)

Subsets and Equality

DEFINITION

A set A is said to be a subset of another set B if every element of A is also an element of B . We use the notation $A \subseteq B$. Two sets A and B are said to be equal (notation $A = B$) if they have the same elements. In other words, A and B are equal if $A \subseteq B$ and $B \subseteq A$.

Subsets and Equality

EXAMPLE

$\{2, 3, 5\} \subseteq \{-1, 2, 3, 5, 7\}$. However, $\{1, 2, 3, 4\} \not\subseteq \{1, 3, 4, 5, 6, 7\}$, because $2 \notin \{1, 3, 4, 5, 6, 7\}$.

EXAMPLE

By definition, $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{R}$.

Set Operations

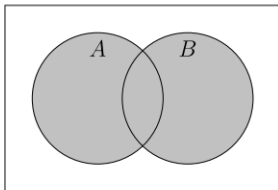
Let A and B be sets.

- ▶ The union of A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or B , or in both. In other words,
 $A \cup B = \{x : x \in A \text{ or } x \in B\}$.
- ▶ The intersection of A and B , denoted by $A \cap B$, is the set that contains those elements in both A and B . In other words,
 $A \cap B = \{x : x \in A \text{ and } x \in B\}$.
- ▶ Two sets are said to be disjoint if their intersection is the empty set.
- ▶ Let S be the universal set (the set of all elements under consideration. It depends on the context). Then the complement A^c of A is defined to be the set of all elements in S that are not in A .

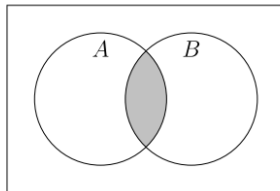
Set Operations, continued

EXAMPLE

Let $A = \{1, 3, 4, 5\}$ and $B = \{1, 2, 3\}$, then $A \cup B = \{1, 2, 3, 4, 5\}$ and $A \cap B = \{1, 3\}$.



Union



Intersection

Set Operations, continued

EXAMPLE

If $P = (-\infty, 2)$ and $Q = [-1, \infty)$, then $P \cap Q = [-1, 2)$ and $P \cup Q = (-\infty, \infty) = \mathbb{R}$.

Set Operations, continued

EXAMPLE

Let $O = \{2k + 1 : k \in \mathbb{Z}\}$ and $E = \{2k : k \in \mathbb{Z}\}$, then O and E are disjoint. If one takes $\mathbb{Z}$ as the universal set, then $O^c = E$ and $E^c = O$.

Set algebra laws

Commutative law:

$$A \cap B = B \cap A$$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

Associative law:

$$(A \cap B) \cap C = A \cap (B \cap C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Distributive law:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Cardinality of the Union of Sets

Let $|S|$ denote the number of elements of a set S . Obviously, $A \cup B$ contains both A and B , so $|A \cup B| \geq |A|$ and $|A \cup B| \geq |B|$. But exactly how many elements are there in $A \cup B$?

THEOREM

Let A, B be sets with finitely many elements, then

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

In particular, if A and B are disjoint, then $|A \cup B| = |A| + |B|$.

Cardinality of the Union of Sets

THEOREM

Let A, B, C be sets with finitely many elements, then

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|.$$

Advanced Set: Algebra

Sets Summary

- ▶ A set is a collection of unique elements. The definition of a specific set determines which elements are members of the set. Elements not specifically defined as members of a set are not in the set.
- ▶ The cardinality of a set is the count of the number of elements in a set based on the sets definition. The cardinality may be finite or infinite.
- ▶ Cardinality example: The cardinality of the set of dwarfs in the Snow White story is 7.
- ▶ The cardinality of the set of integers is NOT the same as the cardinality of the set of real numbers. Both are infinite.

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

Algebra

- ▶ An algebra is a set of elements and a set of laws that apply to the elements.
- ▶ One way to define various types of algebras such as rings, fields, Galois Fields and the like, is to list the possible laws (axioms, postulates, rules) that might apply, then define each algebra in terms of which laws apply.
- ▶ Note: The particular symbols and the particular operations are not important. Addition and multiplication are commonly used for convenience, yet the logical operations "and" and "or" could be used, the set operations union and intersection could be used, as well as many other pairs of operations.

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

Algebra Laws

Closure Laws:

- A0 . Addition is well defined, for every ordered pair $a, b \in S$, $c = a + b$ implies c is a unique element in S .
- M0 Multiplication is well defined, for every ordered pair $a, b \in S$, $c = a * b$ implies c is a unique element in S .

Associative Laws:

- A1 $(a + b) + c = a + (b + c)$ we do not repeat the obvious for all a, b, c in S , after this
- M1 . $(a * b) + c = a * (b * c)$

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

Algebra Laws

Commutative Laws:

A2 Addition $a + b = b + a$

M2 Multiplication $a * b = b * a$

Identity Laws:

A3 0 is in S and $0 + a = a + 0 = a$

M3 1 is in S and $1 * a = a * 1 = a$

Inverse Laws:

A4 for every a in S, -a is a unique element in S and
 $(-a) + a = a + (-a) = 0$

M4 1 is in S and $1 * a = a * 1 = a$

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

Algebra Laws

Commutative Laws:

$$A2 \text{ . Addition } a + b = b + a$$

$$M2 \text{ Multiplication } a * b = b * a$$

Identity Laws:

$$A3 \text{ 0 is in } S \text{ and } 0 + a = a + 0 = a$$

$$M3 \text{ 1 is in } S \text{ and } 1 * a = a * 1 = a$$

Inverse Laws:

$$A4 \text{ for every } a \text{ in } S, -a \text{ is a unique element in } S \text{ and } (-a) + a = a + (-a) = 0$$

$$M4 \text{ for every } a \text{ in } S \text{ except } 0, a ** -1 \text{ is a unique element in } S \text{ and } (a ** -1) * a = a * (a ** -1) = 1$$

Distributive Laws:

$$D1 \text{ } a(b + c) = ab + ac$$

$$D2 \text{ } (b + c)a = ba + ca$$

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

Sets Objects

- ▶ Groups, Rings, and Fields are all sets of elements with additional structure
- ▶ Meaning, various ways of combining elements to produce an element of the set.
- ▶ A GROUP is a set in which you can perform one operation (usually addition or multiplication mod n for us) with some nice properties.
- ▶ A RING is a set equipped with two operations, called addition and multiplication. A RING is a GROUP under addition and satisfies some of the properties of a group for multiplication.
- ▶ A FIELD is a GROUP under both addition and multiplication.
- ▶ $\mathbb{Z}/n\mathbb{Z}$, fancy notation for the integers mod n under addition.
- ▶ $(\mathbb{Z}/n\mathbb{Z})^\times$, more fancy notation for the integers mod n under multiplication.

(ref: The Very Basics of Groups, Rings, and Fields)

Group

- ▶ A group is an algebraic system consisting of a set, an identity element, one operation and its inverse operation.
- ▶ A example group, $G = (S, O, I)$
 - ▶ S is set of integers
 - ▶ O is the operation of addition, the inverse operation is subtraction
 - ▶ I is the identity element zero (0)
- ▶ Another example group, $G = (S, O, I)$
 - ▶ S is set of real numbers excluding zero
 - ▶ O is the operation of multiplication, the inverse operation is division
 - ▶ I is the identity element one (1)
- ▶ The operation does not have to be addition or multiplication.
- ▶ The set does not have to be numeric.

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

Group Axioms

let a , b and c be elements of a group

G1: CLOSURE The operation can be applied to any two elements of the group and the result is an element of the group. $\forall a, b, c \in G, a \circ b = c$

G2: ASSOCIATIVE . $\forall a, b, c \in G, (a \circ b) \circ c = a \circ (b \circ c)$ if operation is addition $(ab)c = a(bc)$ if operation is multiplication

G3: IDENTITY ELEMENT The identity element must be a member of the group and is its own inverse. The identity element is provably unique, there is exactly one identity element. $0 + a = a + 0 = a$ if operation is addition $1a = a1 = a$ if operation is multiplication

G4: INVERSE Every element of the group has an inverse element in the group. $a + (-a) = (-a) + a = 0$ if operation is addition $aa = aa = 1$ if operation is multiplication

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

Groups

DEFINITION

A **GROUP** is a set G which is CLOSED under an operation $*$ (that is, for any $x, y \in G$, $x * y \in G$) and satisfies the following properties:

1. Identity – There is an element e in G , such that for every $x \in G$,
 $e * x = x * e = x$.
2. Inverse – For every x in G there is an element $y \in G$ such that
 $x * y = y * x = e$, where again e is the identity.
3. Associativity – The following identity holds for every $x, y, z \in G$:

$$x * (y * z) = (x * y) * z$$

(ref: The Very Basics of Groups, Rings, and Fields)

Groups

- ▶ A group is said to be “abelian” if $x * y = y * x$ for every $x, y \in G$
- ▶ An Abelian Group or commutative group has an additional axiom
 - ▶ $a + b = b + a$ if the operation is addition
 - ▶ $ab = ba$ if the operation is multiplication
- ▶ A Lie Group is a topological group with only countably many connected components and an identity element that is open.
- ▶ A Cyclic Group is a group that has elements that are all powers of one of its elements.

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

Abelian Groups

A set G with a binary operation $*$ is called an ABELIAN GROUP, if $(G, *)$ satisfies the following axioms:

G1 $*$ is a well-defined map from $G \times G$ to G .

G2 (ASSOCIATIVITY) For all $a, b, c \in G$

$$(a * b) * c = a * (b * c)$$

G3 (EXISTENCE OF A NEUTRAL ELEMENT) There is an element $n \in G$ such that for all $a \in G$

$$a * n = a$$

G4 (EXISTENCE OF INVERSE ELEMENTS) For every $a \in G$ there exists $b \in G$ such that

$$a * b = n$$

G5 (COMMUTATIVITY) For all $a, b \in G$

$$a * b = b * a$$

(Ref: Helmut Knaust Class 200510 3341)

Abelian Groups Examples

- ▶ The sets $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are examples of abelian groups when endowed with the usual addition $+$. The natural element in these cases is 0; it is customary to denote the inverse element of a by $-a$.
- ▶ The sets $\mathbb{Q} \setminus \{0\}$ and $\mathbb{R} \setminus \{0\}$ also form abelian groups under the usual multiplication $\cdot$. In these cases we denote the natural element by 1; the inverse element of a is customarily denoted by $1/a$ or by a^{-1} .

(Ref: Helmut Knaust Class 200510 3341)

Groups Axioms

1. CLOSURE: Given any two integers mod n , their sum (via addition modulo n) is an integer mod n by definition. (Again, to be clear, the operation $*$ described above is addition modulo n .)
2. IDENTITY: $0 \bmod n$ is the identity element, since $a * 0$ means $a + 0 \bmod n$, which is clearly $a \bmod n$.
3. INVERSE: Given any $a \bmod n$, we must find an inverse b so that $a * b = e$ in the group, i.e. $a + b \equiv 0 \pmod{n}$. The inverse to any a in this case is $n - a$.
4. ASSOCIATIVITY: The integers are associative, by basic rules of addition, so the integers mod n are also associative. That is, since

$$a + (b + c) = (a + b) + c, \quad \text{then it follows that} \quad a + (b + c) \equiv (a + b) + c \pmod{n}$$

(ref: The Very Basics of Groups, Rings, and Fields)

Ring

- ▶ A ring is an algebraic system consisting of a set, an identity element, two operations and the inverse operation of the first operation.
- ▶ An example ring, $R = (S, O1, O2, I)$
 - ▶ S is set of real numbers
 - ▶ $O1$ is the operation of addition, the inverse operation is subtraction
 - ▶ $O2$ is the operation of multiplication
 - ▶ I is the identity element zero (0)

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

Rings

DEFINITION

A **RING** is a set R which is CLOSED under two operations $+$ and $\times$ and satisfying the following properties:

1. R is an abelian group under $+$.
2. Associativity of $\times$ – For every $a, b, c \in R$,

$$a \times (b \times c) = (a \times b) \times c$$

3. Distributive Properties – For every $a, b, c \in R$ the following identities hold:

$$a \times (b + c) = (a \times b) + (a \times c)$$

and

$$(b + c) \times a = b \times a + c \times a.$$

(ref: The Very Basics of Groups, Rings, and Fields)

Rings Examples

Examples:

1. Both the examples $\mathbb{Z}/n\mathbb{Z}$ and $\mathbb{Z}$ from before are also RINGS. Note that we don't require multiplicative inverses.
2. $\mathbb{Z}[x]$, fancy notation for all polynomials with integer coefficients. Multiplication and addition is the usual multiplication and addition of polynomials.

(ref: The Very Basics of Groups, Rings, and Fields)

Field

- ▶ A field is an algebraic system consisting of a set, an identity element for each operation, two operations and their respective inverse operations.
- ▶ An example field, $F = (S, O1, O2, I1, I2)$
 - ▶ S is set of real numbers
 - ▶ $O1$ is the operation of addition, the inverse operation is subtraction
 - ▶ $O2$ is the operation of multiplication
 - ▶ $I1$ is the identity element zero (0)
 - ▶ $I2$ is the identity element one (1)

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

Fields

DEFINITION

A **FIELD** is a set F which is closed under two operations $+$ and $\times$ such that

1. F is an abelian group under $+$ and
2. $F - \{0\}$ (the set F without the additive identity 0) is an abelian group under $\times$.

In short, a set F together with an addition $+$ and a multiplication $\cdot$ is called a **FIELD**, if

F1 $(F, +)$ is an abelian group (with neutral element 0).

F2 $(F \setminus \{0\}, \cdot)$ is an abelian group (with neutral element 1).

F3 For all $a, b, c \in F$: $(a + b) \cdot c = (a \cdot c) + (b \cdot c)$.

(ref: The Very Basics of Groups, Rings, and Fields)

Fields Examples

- ▶ The rational numbers and the real numbers are examples of fields.
- ▶ Another example of a field is the set of complex numbers $\mathbb{C}$.

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$$

- ▶ **Examples:** $\mathbb{Z}/p\mathbb{Z}$ is a field, since $\mathbb{Z}/p\mathbb{Z}$ is an additive group and $(\mathbb{Z}/p\mathbb{Z}) - \{0\} = (\mathbb{Z}/p\mathbb{Z})^\times$ is a group under multiplication. Sometimes when we (or Cox) want to emphasize that $\mathbb{Z}/p\mathbb{Z}$ is a field, we use the notation $\mathbb{F}_p$. Other examples: $\mathbb{R}$, the set of real numbers, and $\mathbb{C}$, the set of complex numbers are both infinite fields. So is $\mathbb{Q}$, the set of rational numbers, but not $\mathbb{Z}$, the integers. (What fails?)
- ▶ **Another NON-Example:** If n is not a prime, then $\mathbb{Z}/n\mathbb{Z}$ is not a field, since $(\mathbb{Z}/n\mathbb{Z}) - \{0\} \neq (\mathbb{Z}/n\mathbb{Z})^\times$. There are, in general, lots of other elements than 0 which are not relatively prime to n and hence have no inverse under multiplication.

(ref: The Very Basics of Groups, Rings, and Fields)

Ordered Fields

A field F endowed with a relation $<$ is called an ORDERED FIELD if

O1 For all $x, y, z \in F$

$$x < y \text{ implies } x + z < y + z$$

O2 For all $x, y \in F$ and all $z > 0$

$$x < y \text{ implies } x \cdot z < y \cdot z$$

O3 For all $x, y, z \in F$

$$x < y \text{ and } y < z \text{ implies } x < z$$

O4 For all $x, y \in F$

$$x < y, \quad y < x, \quad \text{or } x = y$$

Both the rational numbers $\mathbb{R}$ and the real numbers $\mathbb{R}$ form ordered fields. The complex numbers $\mathbb{C}$ cannot be ordered in such a way.

(Ref: Helmut Knaust Class 200510 3341)

Ideal

- ▶ An Ideal, I , is a subset of a Ring, R , with the properties:
 - ▶ I is a subgroup of the additive group of R and
 - ▶ for every i in I and every r in R , ir and ri are in I .
- ▶ Example: The set of all multiples of any integer is an Ideal.

(ref: Sets, Groups, Rings and Algebras - CSEE UMBC)

The Completeness Axiom

You probably have seen books entitled “Real Analysis” and “Complex Analysis” in the library. There are no books on “Rational Analysis”.

Why? What is the main difference between the two ordered fields of $\mathbb{Q}$ and $\mathbb{R}$?—The ordered field $\mathbb{R}$ of real numbers is **COMPLETE**: sequences of real numbers have the following property.

Ⓒ Let (a_n) be an increasing sequence of real numbers. If (a_n) is bounded from above, then (a_n) converges.

The ordered field $\mathbb{R}$ of rational numbers, on the other hand, is **not** complete.

The complex numbers $\mathbb{C}$ also form a complete field.

(Ref: Helmut Knaust Class 200510 3341)

Measure Theory

Background

- ▶ Real Analysis and, in particular, measure theory, is very important in probability and statistics.
- ▶ Measure theory itself can be very abstract and difficult. (I am not even skin deep into it)
- ▶ A *measure* is a generalization of the concept of length, area, volume, etc.

Introduction

- ▶ Measure Theoretic Probability offers a very generalized view of probability.
- ▶ Using sets rather than distributions represented by either discrete or continuous functions
- ▶ It allows for complex problems to be understood more simply (if you can get past the rigorous math!)

Measure Theory for Probability: A Very Brief Introduction - Count Bayesie

Measure for Measure

- ▶ Measure Theory is the formal theory of things that are measurable
- ▶ Examples: “How tall are you?”, “What shoe size do you wear?”, “How far to the gas station?”
- ▶ All of these are questions about measuring something in one dimension.
- ▶ “How many square feet is that house?”, “How many acres is the farm?” This is measurement in two dimensions.
- ▶ “How many gallons of milk do you need?”, “How many cubic yards of rocks to fill that hole?” This of course just extends the idea of measurement into three dimensions.

Measure Theory for Probability: A Very Brief Introduction - Count Bayesie

Measure for Measure

- ▶ We also have weight, time, velocity, income, age, etc.
- ▶ All of these can be measured.
- ▶ Sometimes we can come up with things that are NOT measurable!
- ▶ Imagine that you were to build a wall with Lego and then took this wall apart and were somehow able to build two identical walls from only the bricks in the first!
- ▶ If we were trying to take the probability of something, but it turned out to be non-measurable then we would clearly end up in some very strange territory.

Measure Theory for Probability: A Very Brief Introduction - Count Bayesie

Measure for Measure

- ▶ The most basic point of probability is that you are measuring the likelihood of events on a scale from 0 to 1.
- ▶ When we imagine all the things that could happen we're really imagining a 'set' of events.
- ▶ A measurable space is a collection of events B , and the set of all outcomes Ω , which is sometimes called the sample space
- ▶ A measure μ takes a set A (from a measurable collection of sets B), and returns "the measure of A ," which is some positive real number.
 $\mu : B \rightarrow [0, \infty)$.
- ▶ Intuition: when μ is applied to any set, then it gives *Size* of it.
- ▶ Measure of rectangle is $a.b$.
- ▶ Measure of circle is πr^2
- ▶ Measure of entire real line is ?? ∞
- ▶ An example measure is a volume, which goes by the name *Lebesgue measure*

Properties of Measure

- ▶ Nonnegativity: $\mu(A) \geq 0 \forall A \in \mathcal{B}$
- ▶ $\mu(\emptyset) = 0$
- ▶ If A is subset of B then $\mu(A) \leq \mu(B)$
- ▶ Countable Additivity: If $A_i \in \mathcal{B}$ are disjoint sets for $i = 1, 2, \dots$, then the measure of the union of the A_i is equal to the sum of the measures of the A_i .
- ▶ You can see how our ordinary notion of volume satisfies these two properties. Volume cannot be negative and addition of volume is sum (no deduction due to intersection like in sets addition)
- ▶ A probability measure P has the two above properties of a measure but it's also normalized, such that $P(\Omega) = 1$.

A Measure Theory Tutorial (Measure Theory for Dummies) - Maya Gupta

Proofs

(Ref: CS 207 Discrete Mathematics – 2012-2013, IITB, Nutan Limaye <https://www.cse.iitb.ac.in/~nutan/courses/cs207-12/notes/>)

What is a proposition?

A statement that is either true or false.

- ▶ $2 + 2 = 4$, every odd number is a prime, there are no even primes other than 2;
- ▶ $\forall a, b \in \mathbb{N}, \exists c \in \mathbb{N} : a^2 + b^2 = c$;
 $\forall$: for all,
 $\exists$: there exists,
 $\in, \notin$: contained in, and not contained in
 $\mathbb{N}$: the set of natural numbers,
 $\mathbb{Z}$: the set of integers,
 $\mathbb{Q}$: the set of rationals,
 $\mathbb{Z}^+$: the set of positive integers,
 $\mathbb{R}$: the set of reals
- ▶ $\forall a, b \in \mathbb{N}, \exists c \in \mathbb{N} : a^2 - b^2 = c$;
- ▶ $\forall a, b \in \mathbb{N}, \exists c \in \mathbb{Z} : a^2 - b^2 = c$;

It is not always easy to tell whether a proposition is true or false.

Theorems and proofs

THEOREM

If $0 \leq x \leq 2$, then $-x^3 + 4x + 1 > 0$

(scratchpad)

PROOF.

As $-x^3 + 4x = x(4 - x^2)$, which is in fact $x(2 - x)(2 + x)$, the quantity is **positive** non-negative for $0 \leq x \leq 2$. Adding 1 to a non-negative quantity makes it positive. Therefore, the above theorem. □

Theorems and Proofs

Given: a number $n \in \mathbb{N}$

Check: Is n prime?

Why is this algorithm correct?

Is there a number $n \in \mathbb{N}$ s.t

$\forall i : i \in \{2, 3, \dots, \sqrt{n}\} \ i \nmid n,$

but $\exists j > \sqrt{n}$ s.t. $j \mid n$?

Is there a composite $n \in \mathbb{N}$ s.t. all its prime factors are greater than $\sqrt{n}$?

```

1 \FOR{$i=2$ to $\sqrt{n}$}
  \IF{$i \mid n$}
3   \STATE output ‘no’
  \ENDIF
5 \ENDFOR

```

Theorems and Proofs

THEOREM

If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$

PROOF.

As n is a composite, $\exists x, y \in \mathbb{N}, x, y < n : n = xy$. If $x > \sqrt{n}$ and $y > \sqrt{n}$ then $xy > n$. Therefore, one of x or y is less than or equal to $\sqrt{n}$. Say x is smaller than $\sqrt{n}$. It is either a composite or a prime. If it is a prime, then we are done. Else, it has prime factorization (axiom: unique factorization in $\mathbb{N}$) and again, we are done. $\square$

Axioms

Euclid in 300BC invented the method of axioms-and-proofs.

Using only a handful of axioms called Zermelo-Fraenkel and Choice (ZFC) and a few rules of deductions the entire mathematics can be deduced!

Proving theorems starting from ZFC alone is tedious. 20,000+ lines proof for $2 + 2 = 4$

We will assume a whole lot of axioms to prove theorems: all familiar facts from high school math.

Class problems

- ▶ (CW1.1) Prove that for any $n \in \mathbb{N}$, $n(n^2 - 1)(n + 2)$ is divisible by 4. (what about divisible by 8?)
- ▶ (CW1.2) Prove that for any $n \in \mathbb{N}$, $2^n < (n + 2)!$

Bogus proofs

THEOREM (BOGUS)

$$1/8 > 1/4$$

PROOF.

$$3 > 2$$

$$3 \log_{10}(1/2) > 2 \log_{10}(1/2)$$

$$\log_{10}(1/2)^3 > \log_{10}(1/2)^2$$

$$(1/2)^3 > (1/2)^2$$



Another bogus proof

THEOREM

For all non-negative numbers a, b $\frac{a+b}{2} \geq \sqrt{ab}$

PROOF.

$$\frac{a+b}{2} \stackrel{?}{\geq} \sqrt{ab}$$

$$a+b \stackrel{?}{\geq} 2\sqrt{ab}$$

$$a^2 + 2ab + b^2 \stackrel{?}{\geq} 4ab$$

$$a^2 - 2ab + b^2 \stackrel{?}{\geq} 0$$

$$(a-b)^2 \geq 0$$



Proof by contrapositive

THEOREM

*If r is irrational then $\sqrt{r}$ is also irrational.
If $\sqrt{r}$ is rational then r is rational.*

DEFINITION (CONTRAPOZITIVE)

The contrapositive of “if P then Q ” is “if $\neg Q$ then $\neg P$ ”

PROOF.

Suppose $\sqrt{r}$ is rational. Then $\sqrt{r} = p/q$ for $p, q \in \mathbb{Z}$. Therefore,
 $r = p^2/q^2$. □

Proof by contradiction: $\sqrt{2}$ is irrational.

PROOF.

Suppose not. Then there exists $p, q \in \mathbb{Z}$ such that $\sqrt{2} = p/q$, where p, q do not have any common divisors. Therefore, $2q^2 = p^2$, i.e. p^2 is even.

(CW2.1) If p^2 is even, then p is even.

(why?)

Suppose not, i.e. p^2 is even but p is not. Then $p = 2k + 1$ for some integer k . $p^2 = (2k + 1)^2 = 4k^2 + 4k + 1$. As $4(k^2 + k)$ is even, $4k^2 + 4k + 1$ is odd, which is a contradiction.

Therefore, $p = 2k$ for some $k \in \mathbb{Z} \Rightarrow 2q^2 = 4k^2 \Rightarrow q^2 = 2k^2 \Rightarrow q^2$ is even. Therefore, q is even. That is, p, q have a common factor. This leads to a contradiction. □

(CW2.2) Prove that there are infinitely many primes.

Well-ordering principle and Induction

AXIOM (WOP)

Every nonempty set of non-negative integers has a smallest element.

AXIOM (INDUCTION)

Let $P(n)$ be a property of non-negative integers. If

- 1. $P(0)$ is true (Base case)*
- 2. for all $n \geq 0$, $P(n) \Rightarrow P(n+1)$ (Induction step)*

then $P(n)$ is true for for all $n \in \mathbb{N}$.

Well-ordering principle and Induction

AXIOM (STRONG INDUCTION)

Let $P(n)$ be a property of non-negative integers. If

- 1. $P(0)$ is true (Base case)*
- 2. $[\forall k \in \{0, 1, \dots, n\} : P(k)] \Rightarrow P(n+1)$ (Induction step)*

then $P(n)$ is true for for all $n \in \mathbb{N}$.

WOP $\Rightarrow$ Induction

THEOREM

Well-ordering principle implies Induction

PROOF.

Let $P(0)$ be true and for each $n \geq 0$, let $P(n) \Rightarrow P(n+1)$.

Let us assume for the sake of contradiction that $P(n)$ is not true for all positive integers.

Let $C = \{i \mid P(i) \text{ is false}\}$. As C is non-empty and non-negative integers C has a smallest element (due to WOP), say i_0 .

Now, $i_0 \neq 0$. Also $P(i_0 - 1)$ is true, as $i_0 - 1$ is not in C . But $P(i_0 - 1) \Rightarrow P(i_0)$, which is a contradiction. □

THEOREM

WOP $\Leftrightarrow$ Induction $\Leftrightarrow$ Strong Induction [HW]

Using Induction to prove theorems

THEOREM

$$2^n \leq (n+1)!$$

PROOF.

Base case ($n = 0$): $2^0 = 1 = 1!$

Induction hypothesis: $2^n \leq (n+1)!$.

$$\begin{aligned} 2^{n+1} &= 2 \cdot 2^n \\ &\leq 2 \cdot (n+1)! \text{ (by induction hypothesis)} \\ &\leq (n+2) \cdot (n+1)! \\ &\leq (n+2)! \end{aligned}$$



Using Well-ordering principle to prove theorems

Here is a slightly non-trivial example. The following equation does not have any solutions over $\mathbb{N}$:

$$4a^3 + 2b^3 = c^3$$

PROOF.

Suppose (for the sake of contradiction) this has a solution over $\mathbb{N}$.

Let (A, B, C) be the solution with the smallest value of b in S .

(Such an s exists due to WOP.)

Observe that C^3 is even. Therefore, C is even. Say $C = 2\gamma$.

Therefore, $4A^3 + 2B^3 = 8\gamma^3$, i.e. $2A^3 + B^3 = 4\gamma^3$.

Now, B^3 is even and so is B . Say $B = 2\beta$. $\therefore 2A^3 + 8\beta^3 = 4\gamma^3$.

And, now we can repeat the argument with respect to A .

Therefore, if (A, B, C) is a solution then so is (α, β, γ) .

But $\beta < B$, which is a contradiction.



Using Well-ordering principle to prove theorems

It is not always as easy to prove such theorems.

CONJECTURE (EULER, 1769)

There are no positive integer solutions over $\mathbb{Z}$ to the equation:

$$a^4 + b^4 + c^4 = d^4$$

Integer values for a, b, c, d that do satisfy this equation were first discovered in 1986.

It took more two hundred years to prove it.

Recap

- ▶ What are axioms, propositions, theorems, claims and proofs?
- ▶ Various theorems we proved in class:
 - ▶ If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.
 - ▶ If r is irrational then $\sqrt{r}$ is irrational.
 - ▶ $\sqrt{2}$ is irrational.
 - ▶ Well-ordering principle implies induction.
 - ▶ $2^n < n!$
 - ▶ $4a^3 + 2b^3 = c^3$ does not have roots over $\mathbb{N}$.
- ▶ The well ordering principle, induction, and strong induction.

You were asked to think about the following two problems:

- ▶ Is $2^n < \frac{n}{2}!$?
- ▶ For every positive integer n there exists another positive integer k such that n is of the form $9k$, $9k + 1$, or $9k - 1$.

Another proof by induction

THEOREM

For any $n \in \mathbb{N}, n \geq 2$ prove that

$$\sqrt{2\sqrt{3\sqrt{4\ldots\sqrt{n-1}\sqrt{n}}}} < 3 \Leftrightarrow \forall 2 \leq i < j, i, j \in \mathbb{N}, f(i, j) < i + 1 \quad (*)$$

A slightly stronger induction hypothesis is required

PROOF.

For all $2 \leq i \leq j, i, j \in \mathbb{N}$ let $f(i, j) = \sqrt{i\sqrt{i+1}\ldots\sqrt{j}}$.

We will prove a slightly more general statement:

For all $2 \leq i \leq j, i, j \in \mathbb{N}, f(i, j) < i + 1$

This is more general than the theorem statement we wanted to prove.



Another proof by induction

We prove (*) by induction on $j - i$.

Base case: $j - i = 1$. $f(i, i + 1) = \sqrt{i\sqrt{i+1}} < i + 1$.

Induction:

$$\begin{aligned} f(i, j + 1) &= \sqrt{i \cdot f(i + 1, j + 1)} \\ &< \sqrt{i \cdot (i + 2)} && \text{(by Induction Hypothesis)} \\ &\leq i + 1 && \text{(by AM-GM inequality)} \end{aligned}$$

A Bogus Inductive proof

THEOREM (BOGUS, CW2.2)

$a \in \mathbb{R}, a > 0$. Then, $\forall n \in \mathbb{N}, a^n = 1$.

BY STRONG INDUCTION.

Base case: $n = 0$. So $a^n = 1$.

Induction: $n \rightarrow n + 1$.

$$a^{n+1} = \frac{a^n \cdot a^n}{a^{n-1}} = \frac{1 \cdot 1}{1} = 1$$

???



Recap

- ▶ The principle of induction: we proved that $\forall i, j \in \mathbb{N}$, $f(i, j) < i + 1$, where

$$f(i, j) = \sqrt{i \sqrt{i+1} \dots \sqrt{j-1} \sqrt{j}}.$$

- ▶ Take back message: be careful when proving statements by induction.

Mathematical Structures

sets, functions, relations, graphs ...

What are sets?

A set can be vaguely defined as a collection of objects.

But vague definitions can lead to problems.

The barber's dilemma

Once upon a time there was a kingdom in which the king ordered the barber to shave only those who do not shave themselves!

Of course, barber could neither shave himself and nor could he not shave himself!

This is called a paradox.

Why did the paradox arise? – the king should have excluded the barber from the set of all people.

Cantor was the first person to define sets formally – finite sets as well as infinite sets, and prove important properties related to sets.

What are sets?

Let P be a property then he said any collection of objects which satisfy property P is a set, i.e.

$$S = \{x \mid P(x)\}.$$

Russell's paradox:

$$A = \{X \mid X \notin X\}$$

Now if $A \in A$ then $A \notin A$ and if $A \notin A$ then $A \in A$!

(CW) Can you come up with a set that contains itself?

What are sets?

How to get around this paradox?

DEFINITION

Start with a few objects *defined* as sets. Now if A is a set and P is a property, then $S = \{x \in A \mid P(x)\}$ is also a set.

Why does this definition get rid of Russell's paradox?

- ▶ Let $P(x) = x \notin x$. Suppose A is a set and let $S = \{x \in A \mid x \notin x\}$.
 - ▶ ($S \in S$): from the definition of S , $S \in A$ and $S \notin S$, which is a contradiction.
 - ▶ ($S \notin S$): from the definition, either $S \notin A$ or $S \in S$. But we have assumed that $S \notin S$, therefore it must mean $S \notin A$. There is no contradiction!

How to get around Barber's paradox? (CW)

Examples and properties

- ▶ We have already seen sets such as $\mathbb{N}, \mathbb{Z}, \mathbb{R}$ etc.
- ▶ Let A, B be two sets. Their cartesian product, $A \times B$, is defined as

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$
- ▶ Similarly, union, intersection, symmetric difference are defined as:

$$A \cup B = \{x \mid a \in A \text{ or } x \in B\}$$

$$A \cap B = \{x \mid a \in A \text{ and } x \in B\}$$

$$A \oplus B = \{x \mid (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$$

$$\wedge : \text{and}$$

$$\vee : \text{or}$$

Examples and properties

- ▶ Let U be the universe. The complement of a set A with respect to the universe U , denoted as $\bar{A}$ or $A^c = \{x \in U \mid x \notin A\}$.
- ▶ The powerset, $\mathcal{P}(A)$, of a set A is defined to be a collection of all subsets of A .

Example: Let $A = \{a, b\}$ then $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

Infinite sets

- ▶ We have already seen infinite sets:
Examples: $\mathbb{N}, \mathbb{Z}, \mathbb{R}, \mathbb{Q}$.
- ▶ How do we measure the size of any set? For a set S , finite or infinite, $|S|$ denotes the size of that set. It is also called the *cardinality* of the set.
- ▶ For a finite set, $|S|$ equals the number of elements in S .
- ▶ What about infinite sets?
- ▶ Given two infinite sets, can we talk about one being *bigger* than the other? If so, how?

Functions

DEFINITION

Let A, B be two sets. A function from A to B , $f : A \rightarrow B$, is a set defined as follows:

$f = \{(a, b) \mid a \in A, b \in B\}$ with an additional property that if $(a, b) \in f$ and $(a, c) \in f$ then $b = c$.

- ▶ Here, b is called an image of a , denoted as $f(a) = b$.
- ▶ $\text{Range}(f) = \{b \in B \mid \exists a \in A \text{ s.t. } f(a) = b\} \subseteq B$
 $\subseteq$: **Subset of**
- ▶ $\text{Domain}(f) = A$

Types of functions

- ▶ **Injective function, one-to-one:** A function $f : A \rightarrow B$ is said to be injective if $\forall x, y \in A$ if $f(x) = f(y)$ then $x = y$.
 - ▶ Is $f : \mathbb{Z} \rightarrow \mathbb{Z}$, defined as $f(n) = n + 1$ injective?
 - ▶ Is $f : \mathbb{Z} \rightarrow \mathbb{Z}$, defined as $f(n) = n^2$ injective?
 - ▶ What about $f : \mathbb{Z} \rightarrow \mathbb{Z}$, defined as $f(n) = \sqrt{n}$?
- ▶ **Surjective function, onto:** A function $f : A \rightarrow B$ is said to be surjective if $\forall x \in B \exists a \in A$ such that $f(a) = x$.
 - ▶ Is $f : \mathbb{Z} \rightarrow \mathbb{Z}$, defined as $f(n) = n + 1$ surjective?
 - ▶ Is $f : \mathbb{Z} \rightarrow \mathbb{Z}$, defined as $f(n) = n^2$ surjective?
 - ▶ What about $f : \mathbb{R} \rightarrow \mathbb{R}$, defined as $f(x) = 10x - 7$?
- ▶ **Bijective function:** A function is said to be bijective if it is surjective and injective.
 - ▶ Is $f : \mathbb{Z} \rightarrow \mathbb{Z}$, defined as $f(n) = n + 1$ bijective?
 - ▶ Is $f : \mathbb{Z} \rightarrow \mathbb{Z}$, defined as $f(n) = n^2$ bijective?
 - ▶ What about $f : \mathbb{R} \rightarrow \mathbb{R}$, defined as $f(x) = 10x - 7$?

Back to infinite sets

We will understand the notion of *size of an infinite set* in a relative sense.

DEFINITION

We say that two sets A, B have the same size if and only if there is a bijection between A and B

Examples

- ▶ Let E be a set of even numbers. There is a bijection between E and $\mathbb{N}$
 $f(x) = 2x, f : \mathbb{N} \rightarrow E$.
- ▶ There is a bijection $f : \mathbb{Z} \rightarrow \mathbb{N}$
$$f(x) = \begin{cases} -2x & \text{if } x \leq 0 \\ 2x - 1 & \text{otherwise} \end{cases}$$
- ▶ There is a bijection $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$

Thanks ...

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